

Is Blood Transfusion Associated with an Increased Risk of Mortality after Spinal Osteotomy?

A Nationwide Population-based Study

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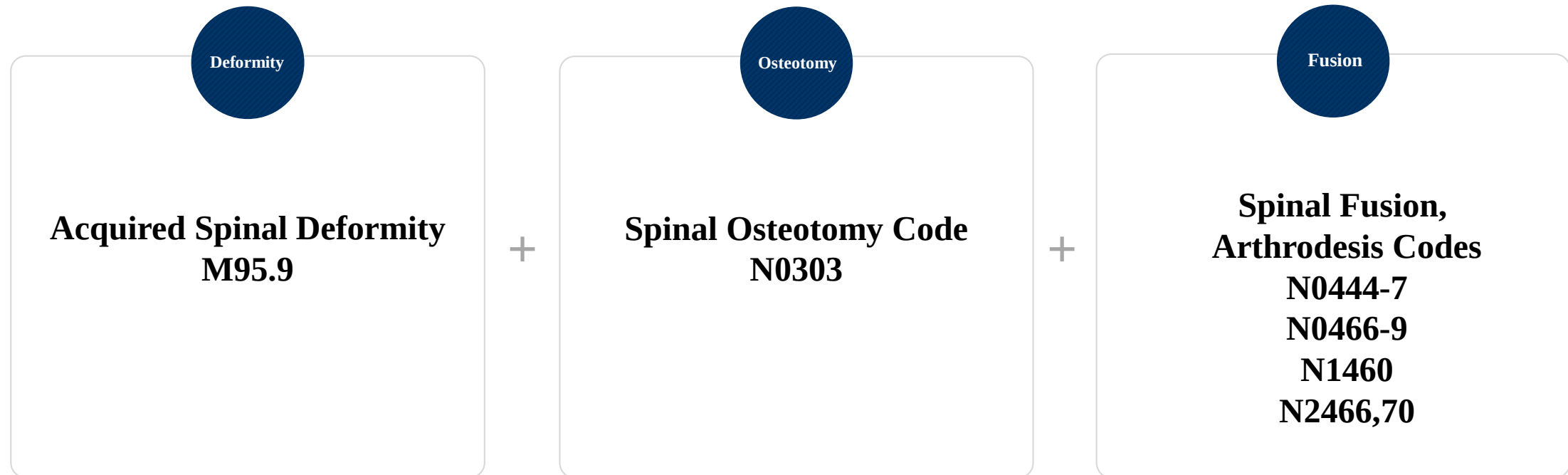
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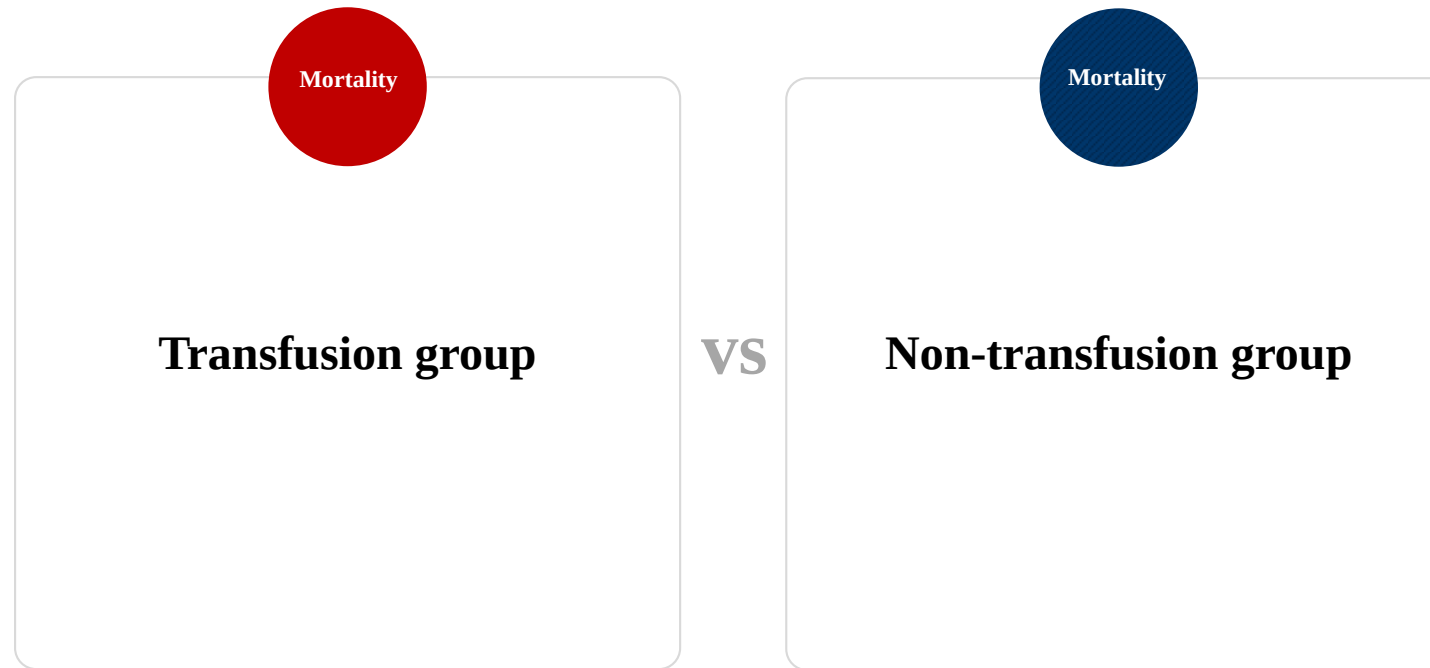
- **Spinal osteotomy (SO) for correcting spinal deformities.**
 - Extensive soft tissue & muscle dissection
- **High blood loss and frequent need for transfusions during SO.**
 - Transfusion-related immunomodulation → poor outcomes & mortality (*Ponnusamy et al. JBJS 2014*)
 - Circulatory overload, pulmonary hypertension (*Zhao et al. NEJM 2016*)
- **To examine the impact of blood transfusions on mortality.**

Korean National Health Insurance Service database (2002-2018)



*Exclusion: previous spine surgery, cancer

Korean National Health Insurance Service database (2002-2018)



Results: Baseline characteristics

- 662 (72.59%) of 912 patients received blood transfusions.
- Average transfusion volume: 1,324 ± 1,120 cc.

Table 1. Baseline characteristics of transfusion and non-transfusion groups.

Variable	Total (n=912)	Transfusion (n=662)	Non-transfusion (n=250)	p-value
Sex, n (%)				
Male	310 (33.99%)	171 (25.83%)	139 (55.6%)	<0.0001
Female	602 (66.01%)	491 (74.17%)	111 (44.4%)	
Age, mean (std)	65.66 (8.84)	66 (8.54)	63.49 (9.25)	<0.0001

- Lower prevalence of lung, kidney, ischemic heart disease in transfusion group.
- No difference in other comorbidities
 - DM
 - Heart failure
 - Cerebrovascular disease
 - Rheumatic disease
 - Liver disease

Table 3. Adjusted hazard ratio of transfusion according to various confounding factors.

Factor		aHR	(95% CI)	p-value†
Female	Sex (ref.: male)	1.506	(1.271–1.785)	<0.0001 ***
	Age (unit: 1 year)	1.009	(1.001–1.018)	0.0373 **
Insurance type (ref.: health insurance)				
	Medical aid	1.063	(0.847–1.333)	0.5988
Hospital size (ref.: tertiary-referral hospital)				
	General hospital	1.022	(0.866–1.205)	0.7962
	Hospital	0.894	(0.724–1.105)	0.2992
	Private clinic	0.716	(0.38–1.348)	0.3007

- Higher short- and mid-term mortality in transfused patients.
- 1-year mortality: **7.25%** in transfusion group vs. **4%** in non-transfusion group. ($p < 0.05$)

Table 4. Mortality rates following spinal osteotomy in Korea between 2002–2018

	SO (n=912)		p-value
≤1 year, 2002~2018	Transfusions (n=662)	Non-transfusions (n=250)	
Deaths, n (%)	48 (7.25%)	10 (4.00%)	0.0426
≤6 months, 2002~2018	Transfusions (n=662)	Non-transfusions (n=250)	
Deaths, n (%)	30 (4.53%)	4 (1.60%)	<0.0001
≤3 months, 2002~2018	Transfusions (n=662)	Non-transfusions (n=250)	
Deaths, n (%)	18 (2.72%)	2 (0.80%)	<0.0001

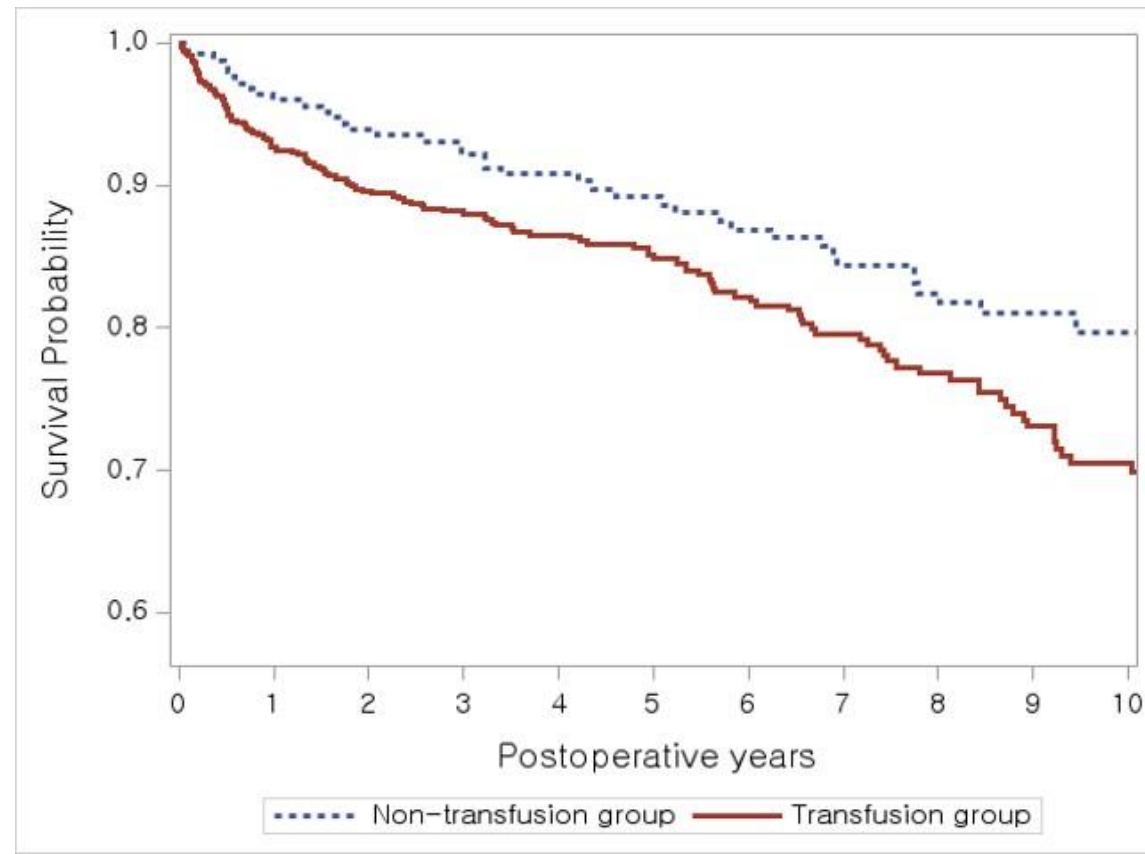
- No significant difference in 10-year mortality.

Table 4. Mortality rates following spinal osteotomy in Korea between 2002–2018

SO (n=912)			p-value
All (total, 2002~2018)	Transfusions (n=662)	Non-transfusions (n=250)	
Deaths, n (%)	140 (21.15%)	44 (17.60%)	0.2341
≤10 years, 2002~2009	Transfusions (n=171)	Non-transfusions (n=65)	
Deaths, n (%)	48 (28.07%)	15 (23.08%)	0.4407
≤5 years, 2002~2014	Transfusions (n=384)	Non-transfusions (n=178)	
Deaths, n (%)	56 (14.58%)	20 (11.24%)	0.2622

Results: Mortality trends over time

- The risk of mortality 48% higher in the transfusion group (aHR 1.476, 95% CI, 1.022-2.13).
- 10-year cumulative postoperative mortality rate higher in the transfusion group.



- With spinal osteotomies, substantial blood transfusions are often needed.
- Patients who receive transfusions exhibit a higher mortality rate.

- Preoperative evaluation and patient blood management to mitigate risks.
 - Female, old age
- Limitations:
 - reliance on insurance data
 - lack of detailed clinical variables (disease severity, osteotomy types, smoking)